2102515 DIGITAL VIDEO PROCESSING · TERM PROJECT

ByteTrack under camera motion

Reproduction and error analysis on MOT17

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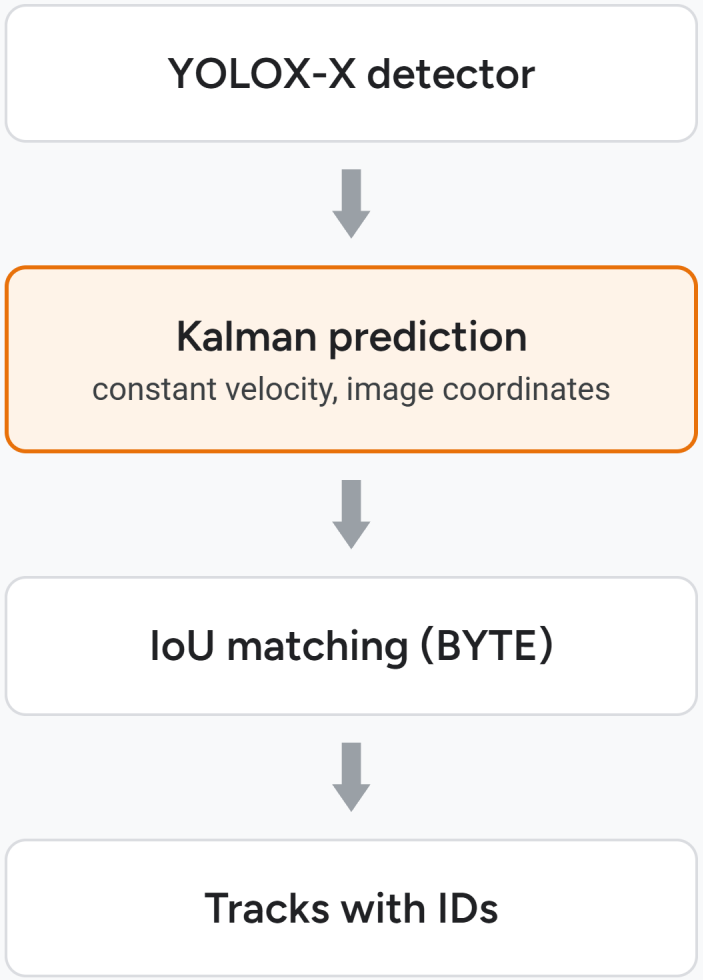
MOTIVATION

A moving camera moves every person in the image

ByteTrack predicts each track with a constant-velocity Kalman filter in **image coordinates**. About half of MOT17 is filmed with a moving camera, but the paper reports only averages.

RESEARCH QUESTION

How much of ByteTrack's error comes from camera motion, and through which part of the tracker?



[1] Zhang et al., ECCV 2022 · [2] Milan et al., arXiv 2016

SETUP

Seven MOT17 sequences, split by camera type

STATIC CAMERA

02 · 04 · 09

fixed viewpoint · 30 fps

MOVING CAMERA

05 · 10 · 11 · 13

hand-held or vehicle · 14–30 fps

Baseline

Official code and ablation weights, default settings

Split

Second half of each sequence: 2,652 frames

Metrics

MOTA, IDF1; HOTA = √(DetA · AssA)

[1] Zhang et al., ECCV 2022 · [2] Milan et al., arXiv 2016 · [5] Luiten et al., IJCV 2021

E1 · REPRODUCTION

Reproduction matches the published numbers

MOTA · paper 76.6

76.6

IDF1 · paper 79.3

79.3

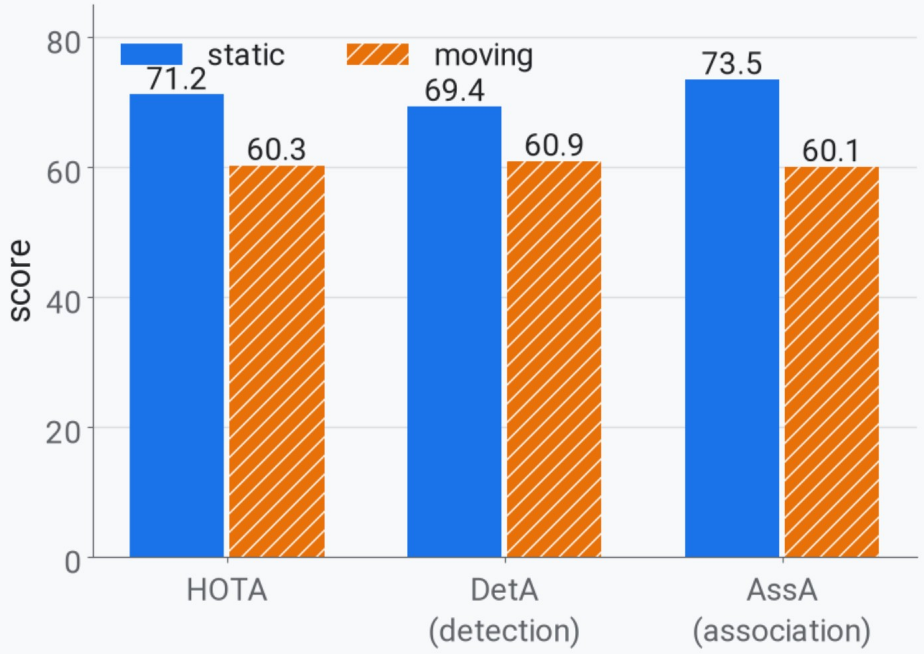
Setting	MOTA	IDF1	IDs
Paper [1]	76.6	79.3	159
Official settings	76.6	79.3	164
+ interpolation (paper: 78.3 / 80.2)	78.3	80.2	148
Matching threshold 0.8	76.4	79.0	202
Detector in FP32	76.6	78.7	167
SORT (paper: 74.6 / 76.9)	74.6	77.2	287

Finding: the published numbers use the code's matching threshold (0.9), not the 0.8 implied by the paper's text.

MOT17 half-val · MOTA/IDF1/IDs with ByteTrack's own evaluation · [1] Zhang et al., ECCV 2022

E2 · STATIC VS MOVING

Moving cameras lose 10.9 HOTA points, mostly in association



31% → 48%
of people vs of identity switches

AssA −13.4
vs DetA −8.4 (detection)

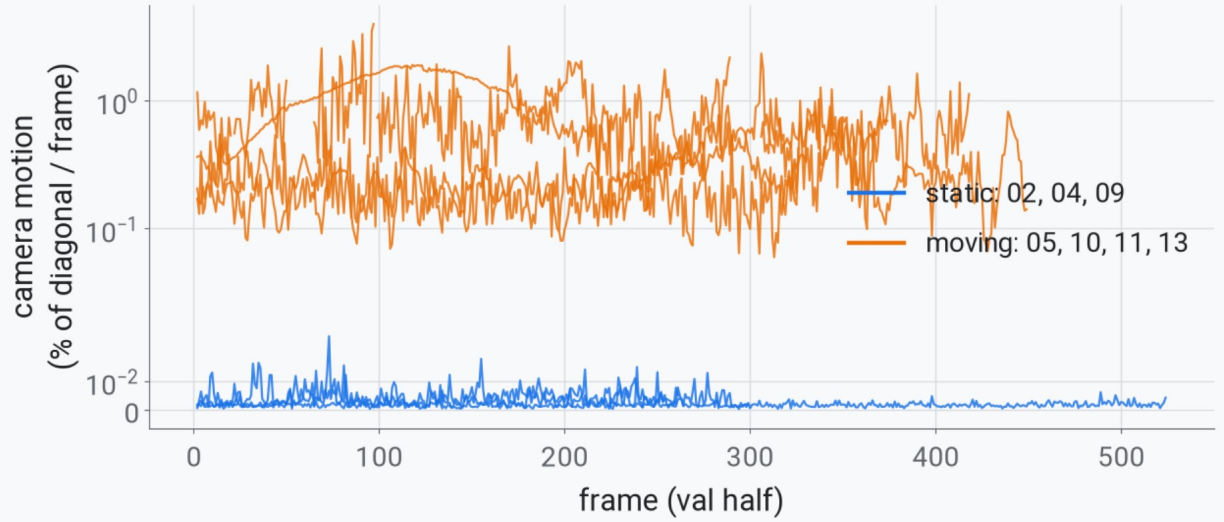
Less crowded, larger people: not crowding

TrackEval, official settings · camera type from [2] Milan et al., arXiv 2016

E3 · MECHANISM

Measuring camera motion per frame

ORB features [3] → people masked → RANSAC fit [4] → corner shift, % diagonal

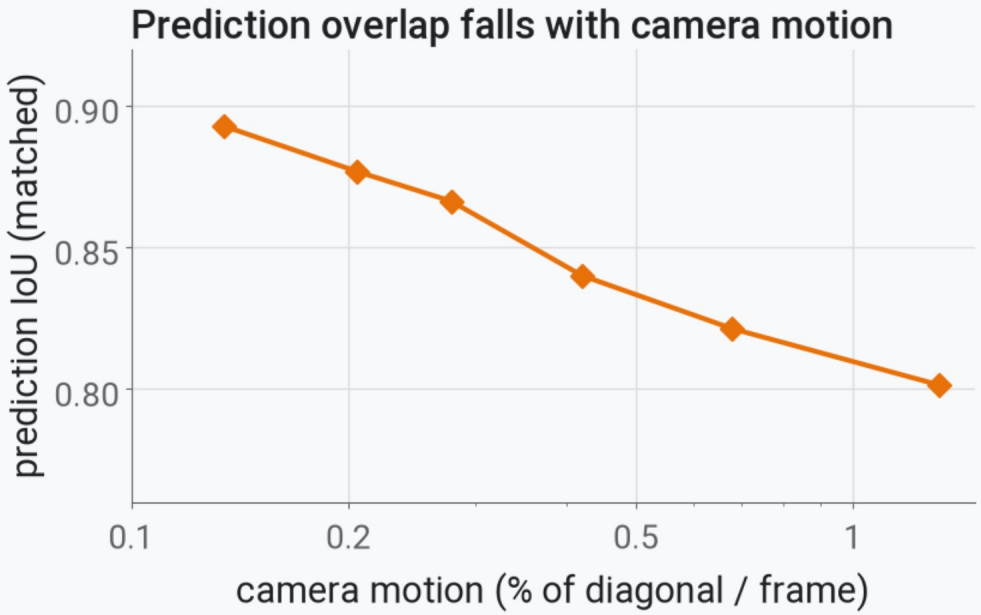


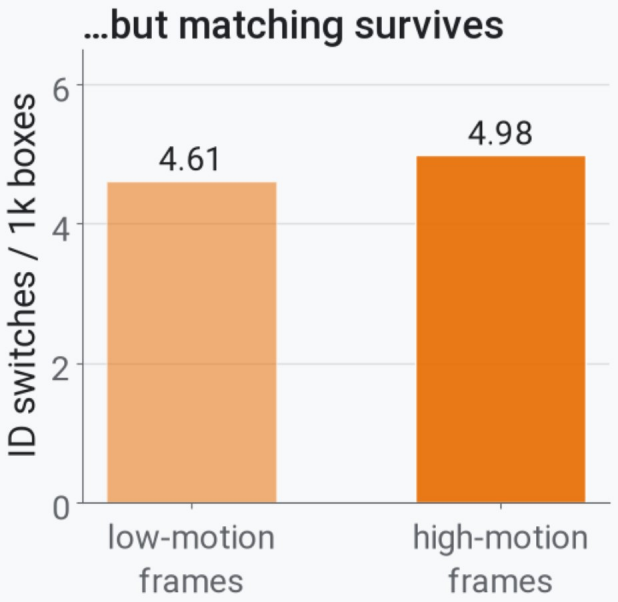
Static ≤ **0.003%** of the diagonal per frame (noise) · moving **0.18–0.70%** · about **100×** more

[3] Rublee et al., ICCV 2011 · [4] Fischler & Bolles, CACM 1981 · Lecture 4: global motion estimation

E3 · ONE-FRAME PREDICTION

Predictions degrade, but matching survives





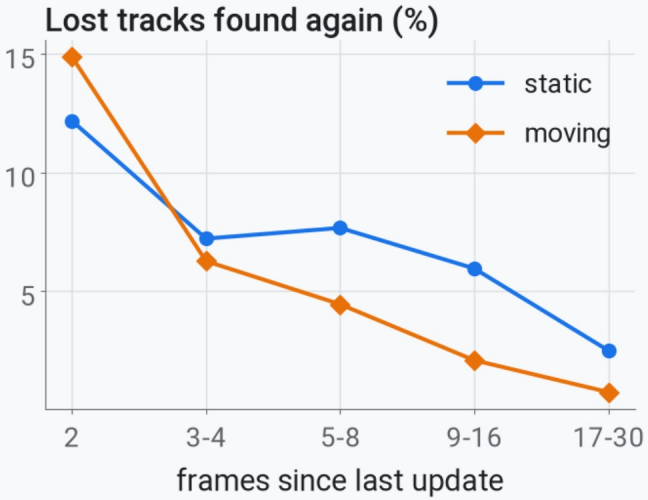
Prediction error grows about **4×** with camera motion (0.011 → 0.047 box heights)

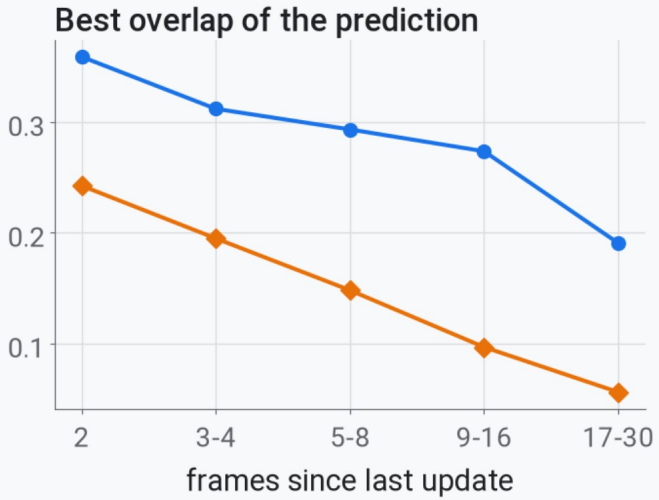
Overlap stays ≈ **0.83**, far above the **0.1** a match needs

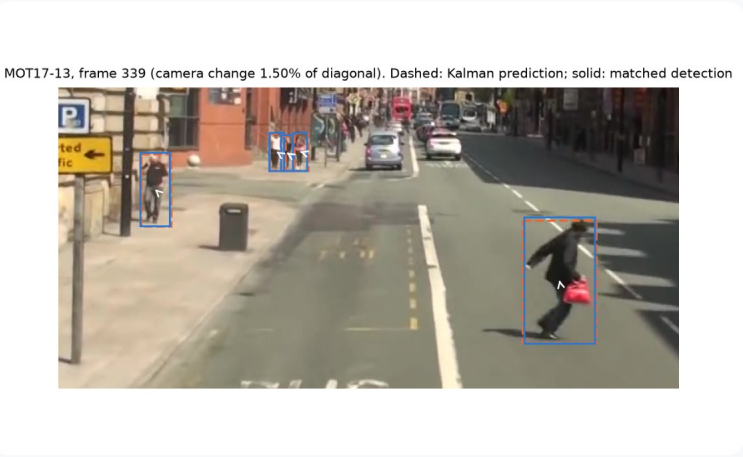
Moving sequences only · within-sequence comparison (frames above vs below each video's median motion)

E3 · LOST TRACKS

Lost tracks are where moving cameras fail







MOT17-13: dashed = Kalman prediction, solid = detection

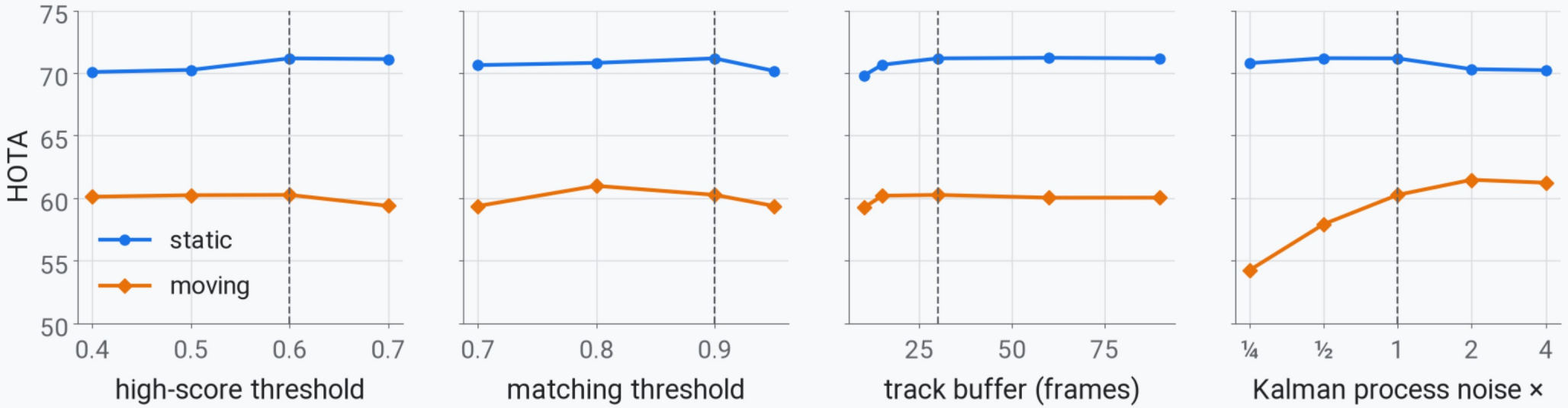
2.1% vs 6.0%
re-found after 9–16 lost frames

New ID
when a track is not found again

ρ = 0.00
extra motion, same gap length

E4 · SENSITIVITY

The gap persists in all 15 settings



8.8–16.5
HOTA gap, never closed

59–61
moving HOTA: thresholds, buffer

54.2 → 61.5
moving HOTA: Kalman noise ×¼ → ×2

One parameter at a time around the official settings · a sensitivity study, not a fix

LIVE DEMONSTRATION

One command, one sequence, one number

```
uv run python scripts/demo.py
```

detect (GPU) → track → evaluate → video

Expected: HOTA 64.8 on MOT17-13, as in the paper's Table II

CONCLUSION

The cost of camera motion lies in recovering lost tracks

1. ByteTrack **reproduced exactly** (76.6 MOTA / 79.3 IDF1); the paper used the code's matching threshold
2. Moving cameras lose **10.9 HOTA**, mostly in association, under all 15 settings tested
3. Camera motion degrades the Kalman prediction, but matching survives; **lost tracks are not re-found**
4. Only the Kalman process noise changes the gap: the **motion model** is the part to study next, e.g. camera-motion compensation [6]

> **Limitations:** seven sequences, correlational evidence; 2-D similarity camera model; one detector and one tracker; one parameter changed at a time

Thank you

Questions are welcome

github.com/vanowarna/bytetrack-moving-camera

REFERENCES

[1] Y. Zhang et al., “ByteTrack: Multi-Object Tracking by Associating Every Detection Box,” ECCV 2022 · arXiv:2110.06864

[2] A. Milan, L. Leal-Taixé, I. Reid, S. Roth and K. Schindler, “MOT16: A Benchmark for Multi-Object Tracking,” 2016 · arXiv:1603.00831

[3] E. Rublee, V. Rabaud, K. Konolige and G. Bradski, “ORB: An Efficient Alternative to SIFT or SURF,” ICCV 2011

[4] M. A. Fischler and R. C. Bolles, “Random Sample Consensus,” Communications of the ACM, 24(6), 1981

[5] J. Luiten et al., “HOTA: A Higher Order Metric for Evaluating Multi-Object Tracking,” IJCV 129, 2021 · arXiv:2009.07736

[6] N. Aharon, R. Orfaig and B.-Z. Bobrovsky, “BoT-SORT: Robust Associations Multi-Pedestrian Tracking,” 2022 · arXiv:2206.14651

Code and results: github.com/vanowarna/bytetrack-moving-camera